

HEMANTHS KUMAR

Core Engineering | Quality Assurance | Hardware Testing | Production Operations
Tumakuru / Bengaluru, Karnataka | +91 97400 29755 | hemanth.1si22ei049@gmail.com
linkedin.com/in/hemanth-s-kumar-207215325 | github.com/HEMANTH-S-KUMAR-1

PROFESSIONAL SUMMARY

Recent Electronics and Instrumentation Engineering graduate (June 2026) from Siddaganga Institute of Technology, Tumakuru, who learns best by building. Hands-on experience comes from two self-driven academic projects, wiring, calibrating, and programming sensor systems with ESP32, HX711 load cells, and Raspberry Pi. Spent five months at Bharat Electronics Limited (BEL) observing defence-grade quality assurance and production documentation on a manufacturing floor, sharpening an eye for process discipline. Picks up C++ and Python quickly for practical, on-the-floor problem solving. Available for immediate hiring.

EDUCATION

B.E. - Electronics and Instrumentation Engineering (EIE) Graduated: June 2026

Siddaganga Institute of Technology (SIT), Tumakuru

CGPA: 6.57 (Semester GPA trending upward: 2nd - 6.00, 4th - 7.00, 6th - 7.83, 8th - 8.69)

Pre-University Certificate (PCMCS) June 2022

Vidyanidhi IND PU College, Tumakuru | Score: 60.33% (362/600)

SSLC (10th Standard) August 2020

Chethana Vidya Mandira HS, Tumakuru | Score: 60.00% (375/625)

PROFESSIONAL EXPOSURE

Engineering Intern - Quality & Operations (Observation & Documentation) Jan 2026 - May 2026

Bharat Electronics Limited (BEL), Bengaluru

- Observed how electronic sub-assemblies are tested and quality-checked inside a strict, defence-grade manufacturing environment.
- Assisted in maintaining documentation covering test parameters, inspection checklists, and component specifications across several production batches.
- Studied PCB inspection procedures and QA workflows used across defence electronics production lines, building a working understanding of formal QA processes in a regulated setting.

HANDS-ON ACADEMIC PROJECTS

Real-Time Safety System in Elevator During Power Failure 2025 - 2026

Siddaganga Institute of Technology

- Built and wired a load cell with HX711 amplifier and Arduino Nano to reliably detect human presence inside a stopped elevator during a power failure.
- Programmed the Arduino Nano to switch to backup power automatically and trigger emergency ventilation (DC fan) and lighting (LED lamp) on failure detection.
- Integrated a GSM module to send real-time SMS alerts to elevator operators with failure details and live passenger status.
- Set up HC-12 wireless transceivers to relay failure data to an external unit, triggering a buzzer and visual alarm for immediate response.

Development of Multichannel Stethograph 2024 - 2025

Siddaganga Institute of Technology

- Interfaced four INMP441 MEMS microphones with an ESP32 via I2S to capture synchronized heart sounds from all four valve locations simultaneously.
- Streamed sensor data from the ESP32 to a Raspberry Pi 4 for real-time signal processing in Python using Librosa, SciPy, and Pydub.
- Built a Tkinter-based GUI to display live heart sound waveforms on an external LCD screen for easy monitoring.
- Extracted time-domain features (Zero Crossing Count, RMS Energy) and frequency-domain parameters (Band Energy Ratio, Spectral Bandwidth) to assess signal quality.

CORE SKILLS

Hardware & Instrumentation: Sensor Assembly & Wiring, Sensor Calibration, Multimeter Troubleshooting, ESP32, HX711 Load Cell, Raspberry Pi, Delta PLC (Programming & Wiring), UART/I2C/SPI

Manufacturing & Quality: QA Workflows, PCB Inspection, Inspection Checklists, Production Batch Documentation, Quality Control, Defence-Grade Standards

Systems & Environments: Linux (Ubuntu), Windows, Industrial Automation, Delta WPL Soft (PLC), MATLAB, LabVIEW, Process Optimization

Programming & Tools: Python - Basic (HackerRank Certified), SQL - Intermediate (HackerRank Certified), C++ (Intermediate), Git, Microsoft Excel (QA Documentation)

CERTIFICATIONS & LEADERSHIP

Certifications: Microsensors and Nanosensors (NPTEL), Electronic Systems for Diagnosis (NPTEL)

Leadership: Volunteer Coordinator, Technisium Technical Fest (2023 & 2024), SIT Tumakuru